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#include <stdio.h>
#include <conio.h>
#define MAX 5
int front=-1,rear=-1;
int q[MAX];

int full()
{
    return((front==0 && rear==MAX-1) || (front==rear+1));
}

int empty()
{
    return(front==-1);
}

void enqueue(int val)
{
    if(full())
    {
        printf("\nqueue overflow");
        return;
    }
    if(front==-1)
    {
        front=0;
    }
    rear=(rear+1)%MAX;
    q[rear]=val;
    // printf("\n %d inserted", val);
}

void dequeue()
{
    if(empty())
    {
        printf("\n queue undeflow");
        return;
    }
    if(front==rear)
    {
        front=rear=-1;
    }
    else
    {
        front=(front+1)%MAX;
    }
}

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    }
}
void display()
{
    if(empty())
    {
        printf("\n queue undeflow");
        return;
    }
    printf("\n queue elemets:");
    int i=front;
    while(1)
    {
        printf("<- %d:%d ->",i,q[i]);
        if(i==rear)
            break;
        i=(i+1)%MAX;
    }
    printf("\n");
}

int main()
{
    int choice ,val;
    while(1)
    {
        printf("\n 1.Enqueue \n 2.Dequeue \n 3.Display \n 4.Exit");
        printf("\n enter choice:");
        scanf("%d",&choice);
        switch(choice)
        {

            case 1:
                printf("\n enter the element to be inserted:");
                scanf("%d",&val);
                enqueue(val);
                break;

            case 2:
                dequeue();
                break;

            case 3:
                display();
                break;

            case 4:
                return 0;
        }
    }
}

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        break;

        default:
            printf("\n invalid choice");
    }
}
return 0;
}
```